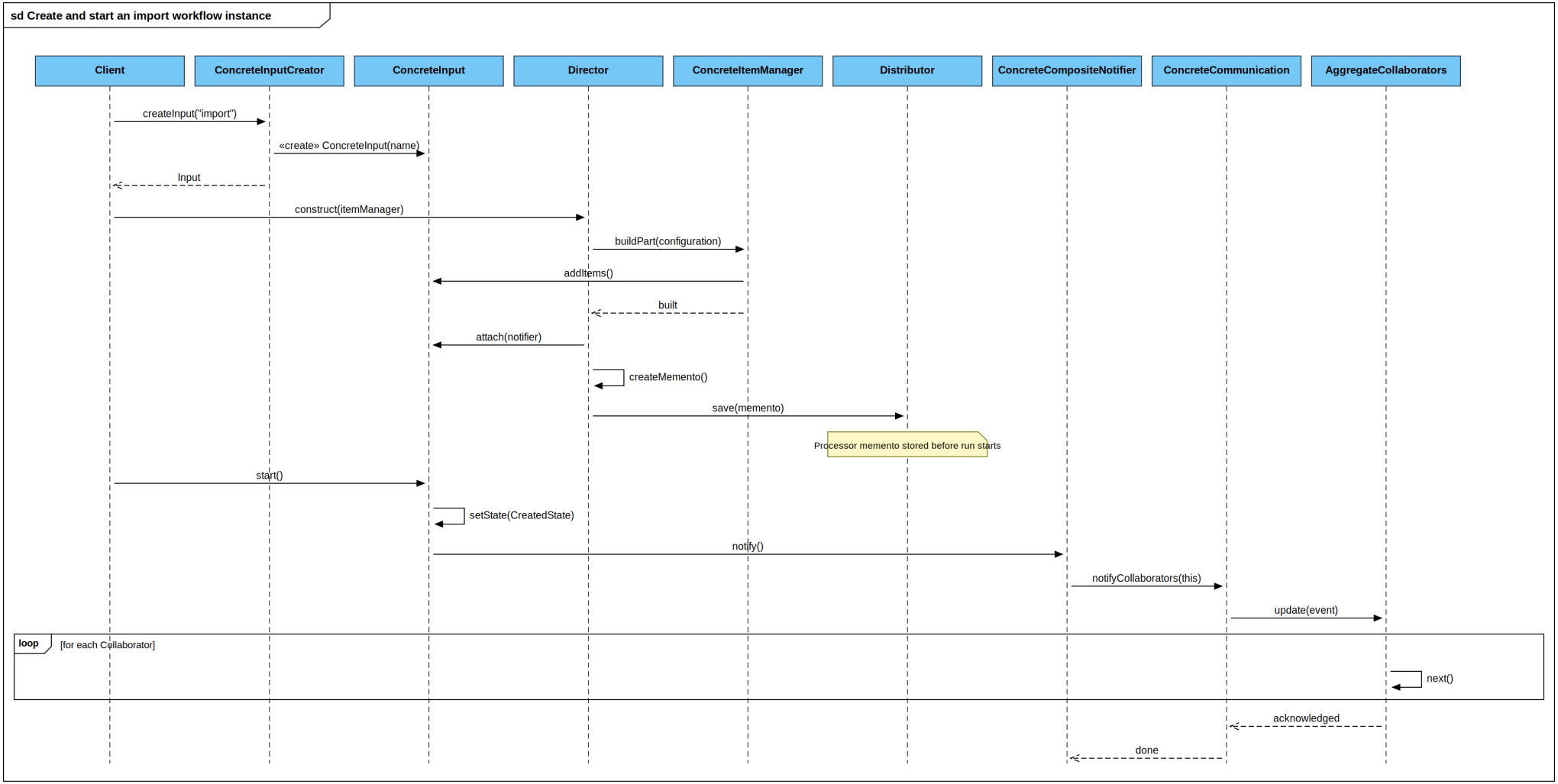
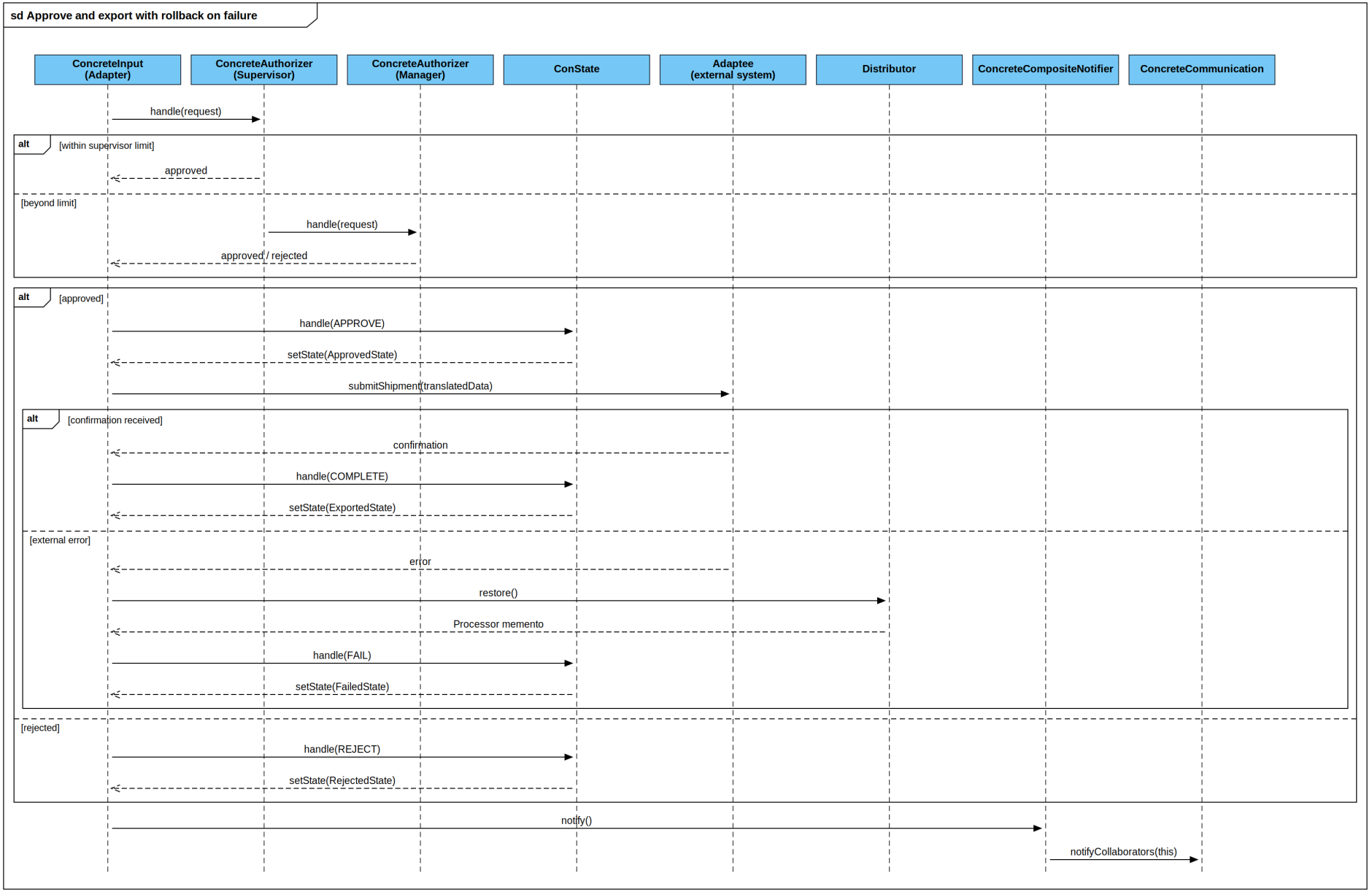
Task 6: Sequence Diagram 1 - Creating and starting an import workflow instance

The client asks ConcreteInputCreator (Factory Method) for an input. Director then drives ConcreteItemManager (Builder) to assemble it, attaches the ConcreteCompositeNotifier (Observer) and has Distributor store a Processor memento (Memento) before the run starts. When the input starts it sets its state (State) and notifies the observer, which passes the event through ConcreteCommunication (Mediator) to AggregateCollaborators, walked with an Iterator.



Task 6: Sequence Diagram 2 - Approval, export and rollback on failure

The request travels along the ConcreteAuthorizer chain (Chain of Responsibility): the supervisor decides within their limit, otherwise the manager. If approved, ConState moves the input to Approved and ConcreteInput, acting as Adapter, submits the shipment to the external system (Adaptee). Success moves the state to Exported. An external error restores the Processor memento via Distributor and moves the state to Failed. A rejection moves it to Rejected. The notifier and mediator then inform the collaborators.



Task 6: State Diagram - Lifecycle of a ConcreteInput (work item)

Each state is a ConState subclass and each transition is a trigger handled through AbstractState. A work item is Created, submitted for approval, then approved, rejected or escalated to a senior authorizer. Approved work runs InProgress and ends Exported on confirmation. An external error moves it to Failed, from where it can retry (memento restored) or be abandoned. Final states: Exported, Rejected and Cancelled.

